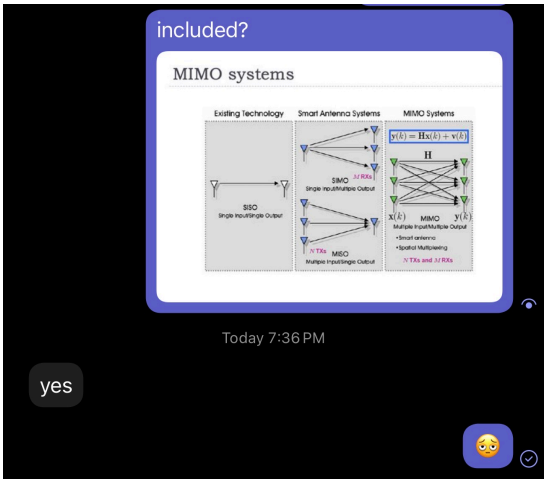
103 - MIMO Systems



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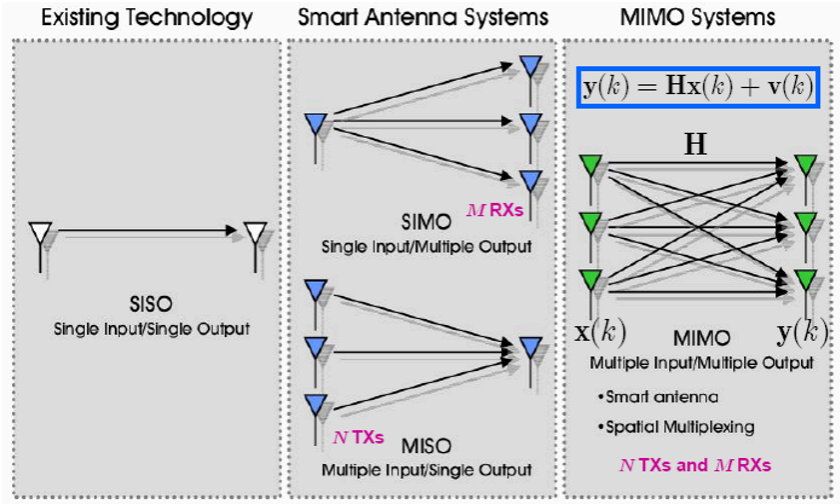
This note is a continuation of 10 - OFDMA.

MIMO Systems



MIMO stands for Multiple Input Multiple Output.

The basic idea is to use multiple transmit antennas and multiple receive antennas instead of just one antenna on each side.

This lets the system improve one or both of these:

- quality of communication, such as better BER
- data rate, by sending more information at the same time

Antenna System Types

The slide compares a few antenna setups:

- SISO = Single Input / Single Output
  - one transmit antenna
  - one receive antenna
  - this is the basic old setup
- SIMO = Single Input / Multiple Output
  - one transmit antenna
  - multiple receive antennas
  - helps mainly at the receiver side
- MISO = Multiple Input / Single Output
  - multiple transmit antennas
  - one receive antenna
  - helps mainly at the transmitter side
- MIMO = Multiple Input / Multiple Output
  - multiple transmit antennas
  - multiple receive antennas
  - gives the most flexibility and the biggest performance gains

The slide also shows the MIMO channel model:

$$y(k) = Hx(k) + v(k)$$

where:

- $x(k)$ = transmitted signal vector at time/sample k
- H = channel matrix between transmit and receive antennas
- $y(k)$ = received signal vector
- $v(k)$ = noise
- with one antenna, there is basically one path from transmitter to receiver
- with many antennas, each transmit antenna can affect each receive antenna
- so the channel becomes a matrix, not just one number

MIMO Basics

- MIMO improves BER / quality and/or data rate
- it does this by using multiple TX/RX antennas

BER = Bit Error Rate, meaning the fraction/probability of transmitted bits that are received incorrectly.

Core Scheme: Space-Time Coding

space-time coding (STC):

The name already tells you the idea:

- space = use multiple antennas
- time = spread or organize symbols across different time instants too

to make transmission more robust or more efficient.

Two Main Functions of STC

1. Diversity

This is about reliability.

The same information is sent in a smarter redundant way across different antennas and/or times so that fading or noise hurts it less.

Goal:

- improve BER
- make the link more robust

2. Multiplexing

This is about rate.

Different data streams are sent in parallel from different antennas.

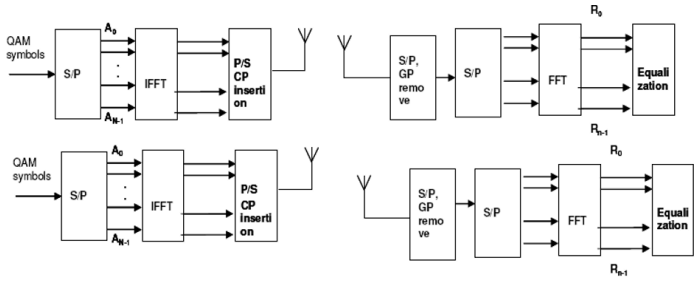
Goal:

- increase bits/sec
- exploit the antenna dimension to carry more data

tl;dr:

- diversity -> better reliability
- multiplexing -> higher throughput

MIMO with OFDMA



MIMO can be combined with OFDMA.

- OFDMA handles sharing resources across subcarriers, time, users, and power
- MIMO uses multiple antennas to improve quality and/or rate

At the transmitter side:

- QAM (Quadrature Amplitude Modulation, recall constellation diagrams) symbols enter
- `S/P` converts serial data to parallel
- `IFFT` creates the OFDM symbol
- cyclic prefix is inserted. Cyclic prefix (CP) is a small copy of the end of an OFDM symbol that is pasted at the beginning before transmission.it helps absorb delayed echoes from multipath and reduce inter-symbol interference (ISI)
- signals are transmitted from multiple antennas

At the receiver side:

- cyclic prefix is removed
- data is converted back into parallel form
- `FFT` recovers the subcarriers
- equalization helps undo the channel effect and remove distortion

Why combine them?

Because the two techniques solve different problems:

- OFDMA is very good for frequency-selective wireless channels and multi-user scheduling
- MIMO adds spatial diversity or spatial multiplexing

So together they can give:

- better robustness
- better throughput
- better overall wireless performance
- OFDMA gives many little frequency lanes
- MIMO gives many antenna paths
- using both means we are using the channel in frequency, time, and space

Multiple Choice Questions

1. What does MIMO stand for?
   A. Multiple Input Multiple Output
   B. Multi-Interval Multi-Operator
   C. Maximum Input Minimum Output
   D. Modulated Input Modulated Output
   Answer: A
2. Which antenna configuration uses one transmit antenna and multiple receive antennas?
   A. SISO
   B. SIMO
   C. MISO
   D. MIMO
   Answer: B
3. What is the main purpose of diversity in MIMO?
   A. Increase routing speed
   B. Improve reliability / BER
   C. Eliminate FFT
   D. Reduce the number of users
   Answer: B
4. What is the main purpose of multiplexing in MIMO?
   A. Increase data rate
   B. Reduce antenna count
   C. Eliminate fading completely
   D. Replace OFDM
   Answer: A
5. In the MIMO channel model $y(k) = Hx(k) + v(k)$, what does H represent?
   A. Header bits
   B. Channel matrix
   C. Hamming distance
   D. Harmonic number
   Answer: B
6. In a MIMO-OFDMA transmitter, which block creates the OFDM symbol?
   A. FFT
   B. IFFT
   C. Equalizer
   D. Router
   Answer: B
7. Why combine OFDMA with MIMO?
   A. Because both do exactly the same job
   B. Because OFDMA uses frequency/time resources while MIMO adds spatial gains
   C. Because MIMO removes the need for subcarriers
   D. Because OFDMA only works in wired networks
   Answer: B

True / False

1. MIMO uses multiple transmit and multiple receive antennas. – True
2. SISO means Single Input / Single Output. – True
3. Space-time coding uses only time and not multiple antennas. – False (space-time coding uses both the antenna dimension and the time dimension)
4. Diversity and multiplexing are two main functions of space-time coding. – True
5. Diversity is mainly used to improve reliability, while multiplexing is mainly used to increase data rate. – True
6. In SIMO, the system has multiple transmit antennas and one receive antenna. – False (SIMO means one transmit antenna and multiple receive antennas)
7. MIMO can be combined with OFDMA to improve performance. – True
8. Wireless channels suffer from both large-scale fading and small-scale fading. – True